

The Quantum Annulus Spectral Constant Problem

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Abstract

Fix $R > 1$. The quantum annulus of type R is the class of invertible Hilbert-space operators T satisfying $\|T\| \leq R$ and $\|T^{-1}\| \leq R$. The associated spectral constant $K(R)$ is the least universal constant for which

$$\|f(T)\| \leq K(R) \max_{R^{-1} \leq |z| \leq R} |f(z)|$$

holds for every operator in this class and every rational function with no pole on the closed annulus. This note gives a self-contained formulation of the problem, its scaling normalization, its scalar, finite-dimensional, and completely bounded variants, the known bounds, and exact criteria for a solution. The sharp value for a fixed finite $R > 1$ remains unknown. The natural candidate is $K(R) = 2$.

1 The annulus and its quantum operator class

For $R > 1$, write

$$\mathbb{A}_R = \{z \in \mathbb{C} : R^{-1} < |z| < R\}, \quad \overline{\mathbb{A}_R} = \{z \in \mathbb{C} : R^{-1} \leq |z| \leq R\}.$$

If $\mathcal{H}$ is a complex Hilbert space, define

$$\mathbb{QA}_R(\mathcal{H}) = \left\{ T \in \mathcal{B}(\mathcal{H}) : T \text{ is invertible, } \|T\| \leq R, \|T^{-1}\| \leq R \right\}. \quad (1)$$

When the underlying Hilbert space is understood, it will be suppressed from the notation.

Every $T \in \mathbb{QA}_R(\mathcal{H})$ satisfies

$$\sigma(T) \subseteq \overline{\mathbb{A}_R}.$$

Indeed, $|\lambda| \leq R$ for $\lambda \in \sigma(T)$, while $\lambda^{-1} \in \sigma(T^{-1})$ gives $|\lambda| \geq R^{-1}$.

The following positive-operator condition is another useful description of the class [4].

Proposition 1.1 (Defect characterization). *For an invertible $T \in \mathcal{B}(\mathcal{H})$, the following are equivalent:*

1. $T \in \mathbb{QA}_R(\mathcal{H})$;

2.

$$\beta_R(T^*, T) := (R^2 + R^{-2})I - T^*T - (T^*T)^{-1} \succeq 0. \quad (2)$$

An invertible operator satisfying $\beta_R(T^*, T) = 0$ is called a *quantum annulus unitary*. Such an operator need not be normal; this terminology should not be confused with a normal operator whose spectrum lies on the boundary of the classical annulus.

1.1 Reduction of a general annulus to the symmetric form

Let $0 < a < b$, and set

$$\mathbb{A}(a, b) = \{z \in \mathbb{C} : a < |z| < b\}.$$

The natural quantum class associated with this annulus is

$$\mathbb{QA}(a, b; \mathcal{H}) = \left\{ T \in \mathcal{B}(\mathcal{H}) : T \text{ is invertible, } \|T\| \leq b, \|T^{-1}\| \leq a^{-1} \right\}.$$

Put

$$c = \sqrt{ab}, \quad R = \sqrt{\frac{b}{a}}, \quad S = c^{-1}T. \quad (3)$$

Then

$$T \in \mathbb{QA}(a, b; \mathcal{H}) \iff S \in \mathbb{QA}_R(\mathcal{H}),$$

and $z \mapsto c^{-1}z$ maps $\mathbb{A}(a, b)$ onto $\mathbb{A}_R$. Moreover, if $g(w) = f(cw)$, then $f(T) = g(S)$ and

$$\max_{a \leq |z| \leq b} |f(z)| = \max_{R^{-1} \leq |w| \leq R} |g(w)|.$$

Consequently, the spectral constant for a general concentric annulus depends only on the ratio b/a , equivalently on its conformal modulus. It is therefore enough to study the symmetric one-parameter family $\mathbb{A}_R$.

Remark 1.2 (A second convention). Some papers use $0 < \rho < 1$ and write the symmetric annulus as $\{\rho < |z| < \rho^{-1}\}$, with $\|\rho T\|, \|\rho T^{-1}\| \leq 1$. This is exactly the convention above with $\rho = R^{-1}$.

2 Spectral and complete spectral constants

For a compact set $X \subseteq \mathbb{C}$, let $\text{Rat}(X)$ be the algebra of rational functions whose poles lie outside X , equipped with

$$\|f\|_X = \max_{z \in X} |f(z)|.$$

If $\sigma(T) \subseteq X$, then $f(T)$ is defined by the rational functional calculus.

Definition 2.1 (K -spectral set). A compact set $X \subseteq \mathbb{C}$ is a K -spectral set for T if

$$\|f(T)\| \leq K \|f\|_X \quad (f \in \text{Rat}(X)).$$

For a fixed $T \in \mathbb{QA}_R(\mathcal{H})$, define

$$\kappa_R(T) = \sup_{\substack{f \in \text{Rat}(\overline{\mathbb{A}_R}) \\ \|f\|_{\mathbb{A}_R} \leq 1}} \|f(T)\|.$$

Definition 2.2 (Universal scalar constant). The spectral constant of the quantum annulus is

$$K(R) = \sup_{\mathcal{H}} \sup_{T \in \mathbb{QA}_R(\mathcal{H})} \kappa_R(T). \quad (4)$$

Equivalently, $K(R)$ is the least constant K such that $\overline{\mathbb{A}_R}$ is a K -spectral set for every $T \in \mathbb{QA}_R(\mathcal{H})$, on every complex Hilbert space $\mathcal{H}$.

There is also a matrix-valued version. If $F = [f_{ij}] \in M_m(\text{Rat}(X))$, define

$$F(T) = [f_{ij}(T)] \quad \text{on } \mathcal{H}^{\oplus m}, \quad \|F\|_X = \max_{z \in X} \|F(z)\|_{M_m}.$$

Definition 2.3 (Universal complete constant). The completely bounded, or complete spectral, constant is

$$K_{\text{cb}}(R) = \sup_{\mathcal{H}} \sup_{T \in \mathbb{Q}\mathbb{A}_R(\mathcal{H})} \sup_{m \geq 1} \sup_{\substack{F \in M_m(\text{Rat}(\overline{\mathbb{A}_R})) \\ \|F\|_{\overline{\mathbb{A}_R}} \leq 1}} \|F(T)\|. \quad (5)$$

Necessarily,

$$K(R) \leq K_{\text{cb}}(R).$$

2.1 Finite-dimensional constants

For $n \geq 1$, set

$$K_n(R) = \sup_{\substack{T \in M_n(\mathbb{C}) \text{ invertible} \\ \|T\|, \|T^{-1}\| \leq R}} \kappa_R(T), \quad K_{\text{fd}}(R) = \sup_{n \geq 1} K_n(R).$$

Then

$$1 \leq K_n(R) \leq K_{\text{fd}}(R) \leq K(R) \leq K_{\text{cb}}(R). \quad (6)$$

The distinction between $K_{\text{fd}}(R)$ and $K(R)$ is substantive: some of the strongest matrix estimates currently use finite-dimensional extremal-function arguments whose validity on an arbitrary Hilbert space is not known.

3 The sharp-constant problem

Problem 3.1 (General quantum annulus spectral constant). For each fixed $R > 1$, determine the exact functions

$$R \mapsto K(R) \quad \text{and} \quad R \mapsto K_{\text{cb}}(R).$$

In particular:

1. determine whether $K(R) = 2$ for every $R > 1$;
2. determine whether $K_{\text{cb}}(R) = 2$ for every $R > 1$;
3. determine whether

$$K_{\text{fd}}(R) = K(R) = K_{\text{cb}}(R);$$

4. identify extremal or asymptotically extremal pairs (T, f) , and decide whether extremality can always be witnessed in finite dimension.

Conjectural sharp value 3.2. For every $R > 1$,

$$K(R) = 2. \quad (7)$$

The stronger complete form asks whether $K_{\text{cb}}(R) = 2$.

Equation (7) is the natural sharp-value candidate because 2 is already a universal lower bound and is the exact limiting value as the annulus becomes thick. It is not currently a theorem for a general fixed finite $R > 1$.

3.1 Equivalent Laurent-polynomial targets

Laurent polynomials are uniformly dense in $\text{Rat}(\overline{\mathbb{A}_R})$. Therefore the scalar assertion $K(R) \leq 2$ is equivalent to the following concrete family of inequalities:

Problem 3.3 (Sharp scalar Laurent inequality). Prove that, for every complex Hilbert space $\mathcal{H}$, every $T \in \mathbb{Q}\mathbb{A}_R(\mathcal{H})$, every $N \geq 0$, and every $a_{-N}, \dots, a_N \in \mathbb{C}$,

$$\left\| \sum_{k=-N}^N a_k T^k \right\| \leq 2 \max_{R^{-1} \leq |z| \leq R} \left| \sum_{k=-N}^N a_k z^k \right|. \quad (8)$$

Likewise, the complete assertion $K_{\text{cb}}(R) \leq 2$ is equivalent to the matrix-valued version:

Problem 3.4 (Sharp matricial Laurent inequality). Prove that, for every $m, N \geq 1$, every $A_{-N}, \dots, A_N \in M_m(\mathbb{C})$, and every $T \in \mathbb{Q}\mathbb{A}_R(\mathcal{H})$,

$$\left\| \sum_{k=-N}^N A_k \otimes T^k \right\| \leq 2 \max_{R^{-1} \leq |z| \leq R} \left\| \sum_{k=-N}^N A_k z^k \right\|_{M_m}. \quad (9)$$

The negative powers in (8)–(9) are essential. A polynomial estimate alone does not solve the annulus problem, because the rational functional calculus must see both boundary components.

4 What is known

Define

$$K_{\text{Tom}}(R) = 2 \left(1 + \frac{2R^2}{(R^2 + 1)\sqrt{R^4 - 1}} \right). \quad (10)$$

Also set

$$K_{\text{BBC}}(R) = 2 + \min \left\{ \frac{R + 1}{\sqrt{R^2 + R + 1}}, \sqrt{\frac{R^2 + 1}{R^2 - 1}} \right\}. \quad (11)$$

Theorem 4.1 (Current universal bounds). *For every $R > 1$,*

$$2 \leq K(R) \leq \min \{ 1 + \sqrt{2}, K_{\text{Tom}}(R) \}, \quad (12)$$

$$2 \leq K_{\text{cb}}(R) \leq \min \{ K_{\text{BBC}}(R), K_{\text{Tom}}(R) \}. \quad (13)$$

Consequently,

$$\lim_{R \rightarrow \infty} K(R) = \lim_{R \rightarrow \infty} K_{\text{cb}}(R) = 2. \quad (14)$$

The ingredients of Theorem 4.1 are:

- Tsikalas's lower bound $K(R) \geq 2$ [8];
- the uniform scalar estimate $K(R) \leq 1 + \sqrt{2}$ of Crouzeix and Greenbaum [2];
- the complete intersection-of-disks estimate $K_{\text{cb}}(R) \leq K_{\text{BBC}}(R) \leq 2 + 2/\sqrt{3}$ of Badea, Beckermann, and Crouzeix [1];

- Tomar's complete estimate $K_{\text{cb}}(R) \leq K_{\text{Tom}}(R)$ [7].

Pascoe first established the exact asymptotic value $\lim_{R \rightarrow \infty} K(R) = 2$ through a degeneration to the quantum cross [6]. The complete estimate (13) now also yields the complete asymptotic statement in (14).

The unresolved scalar gap can be displayed as

$$0 \leq K(R) - 2 \leq \min \left\{ \sqrt{2} - 1, \frac{4R^2}{(R^2 + 1)\sqrt{R^4 - 1}} \right\}. \quad (15)$$

In particular,

$$K_{\text{Tom}}(R) = 2 + \frac{4}{R^2} + O(R^{-4}) \quad (R \rightarrow \infty).$$

Theorem 4.2 (Finite-dimensional bounds). *For matrices of arbitrary finite size,*

$$2 \leq K_{\text{fd}}(R) \leq 1 + \sqrt{1 + \frac{2}{R^2 + 1}}. \quad (16)$$

The lower bound for the all-finite-sizes constant is recorded in [9]; it does not assert that every fixed dimension has constant at least 2 (indeed $K_1(R) = 1$). The upper bound is Crouzeix's finite-dimensional estimate [3]. It is stated there under the strict assumptions $\|T\| < R$ and $\|T^{-1}\| < R$; the non-strict version for rational functions follows by applying the result with $R' > R$, chosen close enough that the function has no pole on $\overline{\mathbb{A}}_{R'}$, and then letting $R' \downarrow R$. The argument producing (16) is not presently known to extend to arbitrary Hilbert-space operators or to the completely bounded setting.

Remark 4.3 (Status). The exact value at a fixed finite modulus remains open; this is stated explicitly in the recent literature [5, 9]. The equality $K(R) = 2$ should therefore be described as the sharp candidate, not as an established result. The quantum cross has exact constant 2, but it is a limiting problem and does not settle any fixed finite R [6].

Remark 4.4 (The degenerate endpoint). If the parameter is formally extended to $R = 1$, the operator class consists of unitaries and the corresponding constant is 1. Thus the lower bound $K(R) \geq 2$ for every $R > 1$ forces a jump at the degenerate endpoint. The behavior as $R \downarrow 1$ from the nondegenerate side is itself an interesting sharp-constant question.

5 Why the problem is not two separate disk inequalities

The two norm assumptions in (1) separately give contraction estimates:

$$\|R^{-1}T\| \leq 1, \quad \|R^{-1}T^{-1}\| \leq 1.$$

Thus the outer disk controls analytic powers of T , and the inner disk controls analytic powers of T^{-1} . A general Laurent polynomial, however, couples these two parts. Intersections of spectral sets need not remain spectral sets with constant 1. The quantum annulus problem asks for the optimal cost of this coupling.

McCullough and Pascoe proved that every $T \in \mathbb{QA}_R$ dilates to a quantum annulus unitary J in such a way that

$$T^n = P_{\mathcal{H}} J^n|_{\mathcal{H}} \quad (n \in \mathbb{Z})$$

[4]. This is a powerful reduction, but J need not be a normal operator supported on $\partial\mathbb{A}_R$. Hence the dilation does not by itself give the constant 1, or determine the sharp constant.

Analytic proofs face the same two-boundary phenomenon. On a convex domain, positivity of the double-layer kernel is a central source of spectral-set bounds. For an annulus, the inner boundary has the opposite orientation and produces an additional term. A sharp proof must preserve enough interaction between the two boundaries to reach the lower bound 2, rather than estimate the two contributions independently.

6 Exact requirements for a complete solution

A proof of the scalar conjecture for a fixed $R > 1$ must establish (8) uniformly in all of the following:

Hilbert space dimension, $T \in \mathbb{QA}_R$, Laurent degree, scalar coefficients.

Because $K(R) \geq 2$, such an upper bound would immediately imply $K(R) = 2$.

A proof restricted to matrices would determine $K_{\text{fd}}(R)$, but an additional approximation, compactness, or dilation argument would be needed to identify that value with the all-Hilbert-space constant. Similarly, a scalar proof does not automatically establish the complete constant: the latter requires the uniform matricial inequality (9).

The following are therefore clean intermediate targets.

Problem 6.1 (Finite-dimensional sharp constant). Determine $K_n(R)$ for each n , and determine

$$K_{\text{fd}}(R) = \sup_{n \geq 1} K_n(R).$$

In particular, decide whether $K_{\text{fd}}(R) = 2$.

Problem 6.2 (Finite-to-infinite transfer). Determine whether

$$K_{\text{fd}}(R) = K(R)$$

for every $R > 1$, and, if so, construct a transfer principle that preserves both bounds $\|T\| \leq R$ and $\|T^{-1}\| \leq R$.

Problem 6.3 (Scalar-to-complete transfer). Determine whether

$$K_{\text{cb}}(R) = K(R)$$

for every $R > 1$. Equivalently, determine whether a sharp scalar annulus inequality automatically amplifies to all matrix levels without loss.

Problem 6.4 (Extremal structure). Classify exact or asymptotic extremizers. In particular, determine whether the lower value 2 can be approached, for fixed R , by matrices of bounded dimension, or whether increasing dimension is intrinsic.

7 A concise master formulation

The complete research problem may be compressed into one statement.

Problem 7.1 (Master problem). Fix $R > 1$. Find the least constants $K_{\text{sc}}(R)$ and $K_{\text{mat}}(R)$ such that, for every complex Hilbert space $\mathcal{H}$, every invertible $T \in \mathcal{B}(\mathcal{H})$ with $\|T\|, \|T^{-1}\| \leq R$, and every Laurent polynomial

$$F(z) = \sum_{k=-N}^N A_k z^k,$$

one has

$$\left\| \sum_{k=-N}^N A_k \otimes T^k \right\| \leq K_{\text{mat}}(R) \max_{R^{-1} \leq |z| \leq R} \|F(z)\|.$$

Here $A_k \in M_m(\mathbb{C})$, uniformly over $m \geq 1$, while the scalar constant $K_{\text{sc}}(R)$ is obtained by restricting to $m = 1$. Thus

$$K_{\text{sc}}(R) = K(R), \quad K_{\text{mat}}(R) = K_{\text{cb}}(R).$$

Determine both constants exactly, decide whether they coincide, and in particular decide whether both are equal to 2.

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